

Autodesk® Scaleform®

Scaleform

Scaleform 4.0

: Vladislav Merker
: 1.01
: 2012 6 19

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFX, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR

IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

:

Scaleform

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

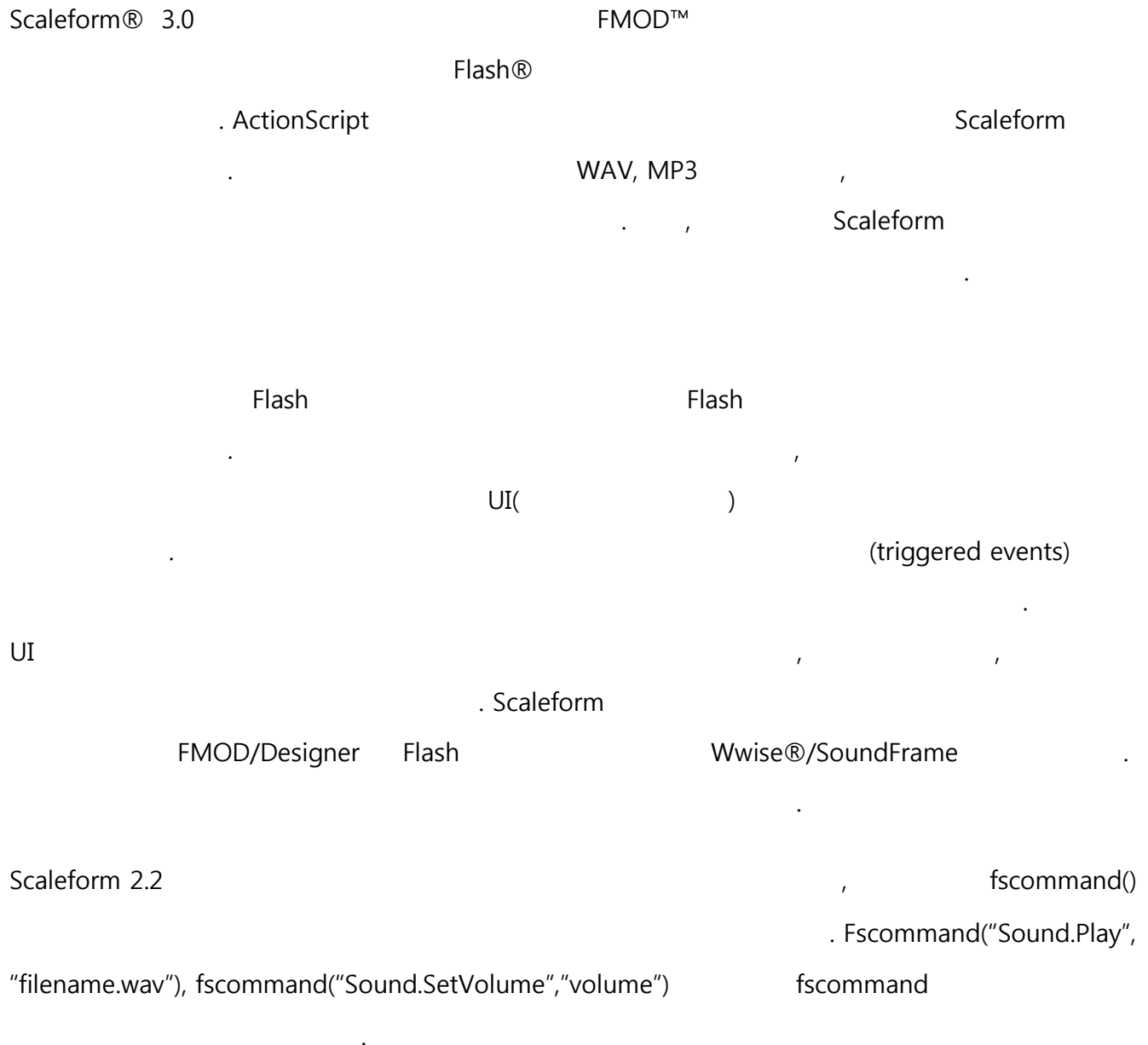
(301) 446-3200

(301) 446-3199

차례

1	소개.....	1
2	사운드 시스템 초기화	2
3	비디오 사운드 시스템 초기화	3
4	사운드 이벤트 인터페이스	4
5	비디오 사운드 시스템 인터페이스.....	5
6	샘플 소스 코드.....	7
7	추가 자료.....	8

1 소개



```

Scaleform                                Gfx::Audio                               Gfx::Loader
      .                                  SWF
Sound::SoundRenderer                      . Sound::SoundRenderer
C++                                       . Scaleform
FMOD                                     .
:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Sound renderer and parameters for playing SWF streaming sounds
    // are provided to the loader by Gfx::Audio. These parameters
    // control when and how the synchronization between SWF frames
    // and sound frames should be done.
    Ptr<Gfx::Audio> pAudioState = *new Gfx::Audio(pSoundRenderer);
    loader.SetAudio(pAudioState);
    pSoundRenderer->SetMasterVolume(SoundVolume);
    pSoundRenderer->Mute(IsMute);
}

Flash
Wwise
Scaleform Player                        Flash
```

3 비디오 사운드 시스템 초기화

```

                                                                    Gfx::Video::VideoSoundSystem
                                                                    . Scaleform
Gfx::Video::Video
Sound::SoundRenderer
:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Setting a video sound system instance which is based on
    // SoundRenderer interface
    pVideo->SetSoundSystem(pSoundRenderer);
}

:
// Setting a video sound system instance which is specific to
// a particular platform
pVideo->SetSoundSystem(
    Ptr<Video::VideoSoundSystem>(*new Video::VideoSoundSystemDX8(0)));

SetSoundSystem()          Gfx::Video::Video
                                                                    .
```

4 사운드 이벤트 인터페이스

Scaleform 3.2 FMOD/Designer Flash
Wwise®/SoundFrame(Sound::SoundEvent::PostEvent Sound::SoundEvent::SetParam)
ActionScript
Scaleform Flash player
Scaleform

1. Sound::SoundEventFMOD - FMOD Ex API
2. Sound::SoundEventFMODDesigner - FMOD EventSystem API , FMOD Designer
3. Sound::SoundEventWwise - Wwise API
4. Sound::SoundEventWwiseSF - Wwise SoundFrame

:

```
#include "Sound/Sound_SoundEventWwise.h"  
Ptr<Sound::SoundEvent> soundEvent = *new Sound::SoundEventWwiseSF;  
  
soundEvent->PostEvent("Sound_PostEvent", "Play_Hello");  
soundEvent->SetParam("Sound_SetParam", "Enable_Effect", 85);
```

Scaleform Player , Flash

5 비디오 사운드 시스템 인터페이스

GFx::Video::VideoSoundSystem GFx::Video::Video

Video::SetSoundSystem()

GFx::Video

- Video::VideoSoundSystemDX8 – Windows DirectSound
- Video::VideoSoundSystemXA2 – Windows Xbox360 XAudio2
- Video::VideoSoundSystemPS3 – PS3 MultiStream
- Video::VideoSoundSystemWii – Wii
- Video::VideoSoundSystemFMOD - FMOD
- Video::VideoSoundSystemWwise – Wwise

Scaleform

Scaleform

Sound::SoundRenderer

GFx::Video::VideoSoundSystem

Video::VideoSoundSystem

Video::VideoSound

Sound::SoundRenderer

Sound::SoundRenderer

Video::Video

.(

.)

:

`#include "Video/Video_VideoSoundSystemXA2.h"`

`pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));`

```

        Video::VideoSoundSystem    Video::VideoSound
        VideoSoundSystem
VideoSoundSystem                    VideoSound
        , Create                    . Scaleform
        . (                          .)    VideoSound
        , Scaleform
        VideoSound::PCMStream
        VideoSoundSystem
        .
        VideoSoundSystemWwise    Wwise SDK    v2009.2.1 build 3271
        . Audiokinetic
        /
        SDK\samples\Plugins\AkAudioInput    Wwise
        . Scaleform Video    Wwise SDK
        .
        :
        #include "Video/Video_VideoSoundSystemWwise.h"
        Ptr<Video::VideoSoundSystem> wwiseSound = Ptr<Video::VideoSoundSystem>(*new
            VideoSoundSystemWwise());
        pVideo->SetSoundSystem(wwiseSound);
        wwiseSound->Update();

        Scaleform Player
        . ,
        Getting Started with Video
    
```

6 예제 소스 코드

- Scaleform Player , FMOD/Wwise :
 - o Apps\Samples\FxPlayer\FxPlayer.cpp
 - o Apps\Samples\Common\FxSoundFMOD.cpp
 - o Apps\Samples\Common\FxSoundWwise.cpp
- FMOD :
 - o Src\Sound\Sound_SoundRendererFMOD.h
 - o Src\Sound\Sound_SoundRendererFMOD.cpp
- (FMOD/Designer Wwise/SoundFrame):
 - o Src\Sound\Sound_SoundEventFMOD.h
 - o Src\Sound\Sound_SoundEventWwise.h
- :
 - o Src\Video\Video_VideoSoundSystem*.cpp
- Flash :
 - o Bin\Data\AS2\Samples\Audio_Demo_AS2 fla
 - o Bin\Data\AS3\Samples\AudioDemo_AS3 fla

7 추가 자료

CRI Movie ,
(export)

ActionScript

SWF

- [Adobe Flash ActionScript 2.0](#)
- [Scaleform - Flash Support Overview](#)
- [Getting Started with Video](#)
- [CRI Movie Encoder Command Line Tool Overview](#)
- [GFxExport Reference Guide](#)